
What does self-supervision add to engineered features for stellar light curves?

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Abstract

Self-supervised models promise one representation, pre-trained once on unlabelled survey data and read out cheaply for many downstream tasks. In stellar time-domain astronomy that promise has to improve on a strong, cheap incumbent: a few dozen engineered light-curve statistics. We pre-train a Conv1D variational encoder with a recurrent latent-dynamics model on the unlabelled TESS 2-minute light curves of 9,428 stars, freeze it, and ask whether the features it learns add anything to 25 engineered features under one linear readout, on ten labelled tasks spanning variability classification, asteroseismology and rotation. Concatenating the two improves the readout beyond 2 SE on 7 of 10 tasks; an untrained encoder does not. Against two networks trained end-to-end on the same labelled stars, splits and inputs, the combined readout is ahead of a supervised Conv1D with the encoder's architecture on 6 of 10 tasks and indistinguishable on the other 4, and ahead of an MLP on raw flux on 9 of 10. **Self-supervision adds to engineered features rather than replacing them. Under a fixed linear readout, the combination matches or exceeds the two supervised networks we train on the same 669 to 16,002 labelled stars per task.**

1 Introduction

Wide-field photometric surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] trades sampling rate against sky coverage: 2-minute (or 20-second) postage stamps for hundreds of thousands of pre-selected stars, and full-frame images, now every 200 s, for every star in the field. LSST [Ivezić et al., 2019] will enlarge the gap by orders of magnitude. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each downstream task. Encoders of this shape exist for ground-based and Kepler light curves [Donoso-Oliva et al., 2025, Zuo et al., 2025, Rui, 2026] and for TESS [Poznanski, 2026]; they are validated by downstream accuracy on their own, or by the stability of the embedding itself, and none measures what the representation adds to the engineered features.

What makes this domain unusual is the strength of the incumbent it must improve on. Decades of variable-star research have distilled light curves into compact engineered summaries: periodogram peak structure, band-limited power, spectral entropy and robust scatter statistics. Pipelines built on them remain highly competitive [Audenaert et al., 2021]. The benchmark of Li et al. [2025] makes the point sharply: across seven expert-labelled variable-star classes, no time-series foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned representation for this data therefore has to answer a specific question: does it carry information the engineered features do not already have?

We measure that directly. We pre-train a latent-dynamics encoder on unlabelled TESS photometry, freeze it, and ask what the features it learns add to a 25-feature engineered stack under an identical

linear readout, and whether the resulting pipeline, which trains only a linear model on the labels, matches or exceeds two networks trained end-to-end on the same labels. Our contributions are:

- **The learned features add to the engineered features rather than replacing them.** Concatenating the two improves a fixed linear readout beyond 2 SE on 7 of 10 tasks, with the gains concentrated where the engineered basis is information-starved; an untrained-encoder control shows the effect is not generic column addition.
- **The combined pipeline, read out by a fixed linear model, matches or exceeds the two supervised networks we train end-to-end on the same labels.** Against a supervised Conv1D with the encoder’s architecture it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

2 Data and method

Data. We use 2-minute SPOC PDCSAP light curves of stars brighter than TESS magnitude $T = 10$ from Sectors 27–101. Pre-training and five of the ten probes (pulsating, eclipsing binary, rotation, transit and rotation period) use a label-enriched subset of 13,470 stars, split 70/15/15 by star: every star labelled as a transit host, eclipsing binary or pulsator that has a usable window, plus a random sample of 10,000 stars matched in no variability catalogue. The pre-training loss never sees a label, but this pool is label-enriched, not a random draw from the survey. Each light curve is split into contiguous segments wherever the time axis breaks, at a missing cadence or a jump longer than five median cadences; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences ($256 \times 120 \text{ s} \approx 8.5 \text{ h}$); restricting to one cadence keeps every window the same span of time. Windows containing a missing value are discarded, never interpolated or padded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of exactly 16 consecutive windows ($\approx 5.69 \text{ d}$), cut at a random offset within a segment, and uses no labels of any kind.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The training objective is

$$\mathcal{L} = \|\hat{x} - x\|^2 + \beta \text{KL} + \lambda [\|\hat{z}_{t+1} - \bar{z}_{t+1}\|^2 + \|\hat{z}_{t-1} - \bar{z}_{t-1}\|^2] + w [\|\log P_H(\hat{x}) - \log P_H(x)\|^2 + \|\Delta\hat{x} - \Delta x\|^2],$$

where every term is a mean squared error, $\bar{z}$ the stop-gradient target of the forward and backward predictors, P_H the Hann-tapered power spectrum, Δ the first difference, and $(\beta, \lambda, w) = (0.1, 60, 0.3)$, β warmed up over ten epochs. The encoder is trained with six seeds and then frozen; no gradient from any label reaches it.

Representations and readout. A star’s learned features are μ , the mean, over every window of its first segment (16 for 95% of stars, 20 for the rest), of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021], computed from one periodogram and the flux distribution of the same first segment; Appendix D defines each feature. Readouts are fixed in advance and never tuned on a task, standardised on the training split alone: class-balanced L_2 logistic regression at $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation (cross-validating C changes no gain by more than 0.011). We compare four arms, each the same readout on a different feature set: features, μ , features $\oplus \mu$, and, as a control, the same fusion built on an *untrained* encoder; the two supervised baselines below are the fifth and sixth arms.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk unchanged, adds a 128-unit linear layer and a scalar output, and is trained per task on a star-level score, the mean of the star’s window logits computed inside the network so that the gradient flows through the mean, with class weighting. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so C1 versus C2 isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with six seeds.

Table 1: Ten-task scorecard. Linear-readout arms use a fixed logistic or ridge readout; the two supervised arms are trained end-to-end per task. Bold marks a leader separated from the runner-up by the combined 2 SE band; italic marks a leader that is only the point-estimate argmax. n is the test-star count and prev. the positive-class prevalence.

task	metric	n (prev.)	frozen linear readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	<i>0.775</i>	0.774	0.719
rotation	PR-AUC	2021 (0.089)	0.540	0.559	<i>0.555</i>	<i>0.565</i>	0.526
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	<i>0.396</i>	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.832	0.834
rotation period	R^2	150	0.703	0.677	0.717	0.597	0.698
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	<i>0.533</i>	0.444

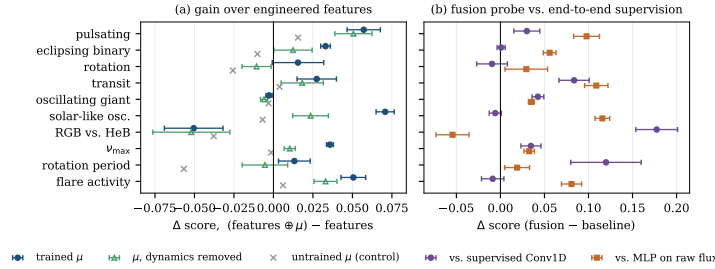


Figure 1: The two claims. **(a)** What the learned features add to the engineered features under the same fixed linear readout, for the shipped encoder, its dynamics-free twin and an untrained control. **(b)** The fusion probe against the two supervised baselines. Bars are 2 SE; the untrained control is a single encoder and carries none.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by R^2 . The other five tasks draw their labels from catalogues that barely overlap the pre-training subset and are scored on a second pool of 22,860 stars built the same way (every catalogue positive with a usable window plus 10,000 quiet stars, split 70/15/15); 1,471 of them were in the pre-training split, so these tasks mostly score stars the encoder never saw.

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six); the engineered arm is deterministic. A μ -arm seed indexes a separate pre-training run and a C1 or C2 seed only an initialisation, so seed k of one has nothing in common with seed k of the other; such differences are treated as unpaired and their bands combined as $\sqrt{SE_a^2 + SE_b^2}$. A difference counts as resolved only when it exceeds the combined 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, seeds averaged; Appendix C) checks that no verdict hinges on the particular test stars.

3 Results

The learned features add to the engineered features. Table 1 and Figure 1a give the comparison. Concatenating μ with the 25 features improves the fixed readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators +0.071, pulsating +0.057 and flare activity +0.051, and reaching +0.013

(R^2) on rotation period. Rotation is positive but just short of the bar (+0.016 against 0.016). The two negative tasks are the ones where 128 extra columns cost more than they add: the engineered basis is already saturated (oscillating giants, -0.003 on a baseline of 0.920) or the training set is too small to pay for them (RGB versus helium-burning, -0.051 on 755 training stars). On both, an untrained-encoder control loses by an amount we cannot resolve from the trained fusion’s, the signature of collinear columns diluting a linear fit, not of μ carrying misleading content (Appendix A). Across all ten tasks the untrained control is negative on seven and never exceeds +0.016, so the gains are not what any 128 extra columns would buy on this feature set; they are specific to what pre-training learned. μ on its own does not replace the features: it is ahead on only 4 of 10 tasks alone, which is the conclusion Li et al. [2025] reach for frozen time-series foundation models, and is why we report fusion rather than substitution [Makinen et al., 2024].

The fusion probe matches or exceeds the two supervised networks. Against C1, which shares the convolutional trunk, labels and input windows but is trained end-to-end, the fusion probe is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it clears the engineered arm on 5 of 10 and the μ -only arm on 4, and on seven tasks the fusion probe recovers 0.98–1.09 of its margin above the metric floor. A paired star-bootstrap (Appendix C) excludes zero on 5 of the 7 fusion gains, 5 of the 6 leads over C1 and 7 of the 9 over C2; the rest stay ahead in at least 65% of resamples and none changes sign. This is a statement about this architecture on 669 to 16,002 labelled stars per task, not about supervision in general.

The latent-dynamics term is what buys the complementarity. Replacing the forward–backward dynamics objective with a dynamics-free variant, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Figure 1a, open triangles; largest losses solar-like oscillators -0.047 , rotation -0.026 , $\nu_{\max}$ -0.025). The prediction term, not the autoencoder alone, supplies what the engineered features lack.

4 Discussion

Model selection must be gated on a held-out probe. The reconstruction objective develops a localised artifact at the window boundary, and none of eleven interventions satisfied both pre-registered acceptance criteria, so the tapered variant shipped on the stated rule, not because it won. Underlying this is one measurement (Appendix B): across ten pre-training recipes of one sweep, validation reconstruction *anti*-predicts downstream probe score: the Spearman ρ between negated validation reconstruction and PR-AUC runs from -0.58 to -0.84 over four tasks, the loss-optimal recipe is the dynamics-free one on all four, and trusting the loss costs 0.021–0.049 PR-AUC. In the recipe family we ship the sign is mixed (ρ from -0.51 to $+0.37$): there the loss does not predict the probe, and in neither family does it pick the probe-optimal recipe. Across seeds of one recipe no label-free metric we tried ranks anything. Self-supervised representation work here therefore cannot be tuned on its own objective and must be gated on a held-out probe.

Limitations and next steps. The four linear-readout arms (features, μ , features $\oplus \mu$, and the untrained control) share one linear model. It has few parameters, so it cannot rescue weak features, and it is identical across the four, so the comparison is between feature sets. Our claim is confined to these 25 features and a linear readout. Every arm, the supervised baselines included, sees one 5.7-day segment of a bright 2-minute star. The fainter, far more numerous stars in the full-frame images, and light curves spanning several sectors, are the obvious next experiments. Our negatives are stars absent from the label catalogues, not stars verified to lack the signal, so every detection score is a lower bound. The flare label marks stars that have flared somewhere in their TESS data, not a flare inside the 5.7-day segment we encode, so the task is recognising a flaring star, not finding a flare. When the readout is trained on one percent of the labels, the advantage narrows on 6 of 10 tasks, so how much pre-training helps when labels are scarce remains an open question.

Reproducibility and disclosure. §2 and Appendix D give the data, objective, features and readouts; the encoder has four convolutional stages (widths 32/64/128/256, kernel 5, each halving length), a 128-d latent and a one-layer GRU of width 256. Code and score tables will be released on publication; generative AI assisted with analysis code and prose editing.

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213 A The two tasks where fusion does not help

task	features	fusion Δ	2 SE	untrained Δ	n_{test}
oscillating giants	0.920	−0.003	0.002	−0.003	3,429
RGB vs. He-burning	0.758	−0.051	0.019	−0.038	161

215 **Oscillating giants: the engineered basis is already near-sufficient.** The labels of Hon et al. [2021]
 216 come from a convolutional classifier reading a 128×128 image of each star’s power spectrum, which
 217 fires on a visible oscillation power excess at $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$. Our engineered vector encodes
 218 close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral
 219 centroid and entropy, locate the excess and describe its concentration. The features accordingly reach
 220 0.920, and the trained and untrained fusion arms then lose the same 0.003: 128 collinear columns
 221 diluting a linear fit, with no trained-versus-untrained difference left to attribute to μ .

222 **RGB versus helium-burning: no arm resolves the discriminant.** Sreenivas et al. [2026] assign the
 223 evolutionary state with a convolutional classifier reading the background-corrected power spectrum
 224 *folded at $\Delta\nu$* ; the physics underneath is the dipole mixed-mode period spacing $\Delta\Pi_1$ [Bedding
 225 et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window
 226 at a time, so the finest frequency structure available to μ is $\Delta f = 32.6 \mu\text{Hz}$, while $\Delta\nu$ for these
 227 stars is $\approx 1.5\text{--}18 \mu\text{Hz}$; the comb itself is unresolved, let alone a fold of it. The engineered features,
 228 computed on the whole 5.69 d segment, reach $2.0 \mu\text{Hz}$, still coarser than the mixed-mode spacing
 229 $\Delta\Pi_1\nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$ near $\nu_{\text{max}} = 50 \mu\text{Hz}$. No arm on this row measures the quantity the label
 230 encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its
 231 discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the fusion
 232 probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the
 233 study.

B Validation reconstruction against downstream score

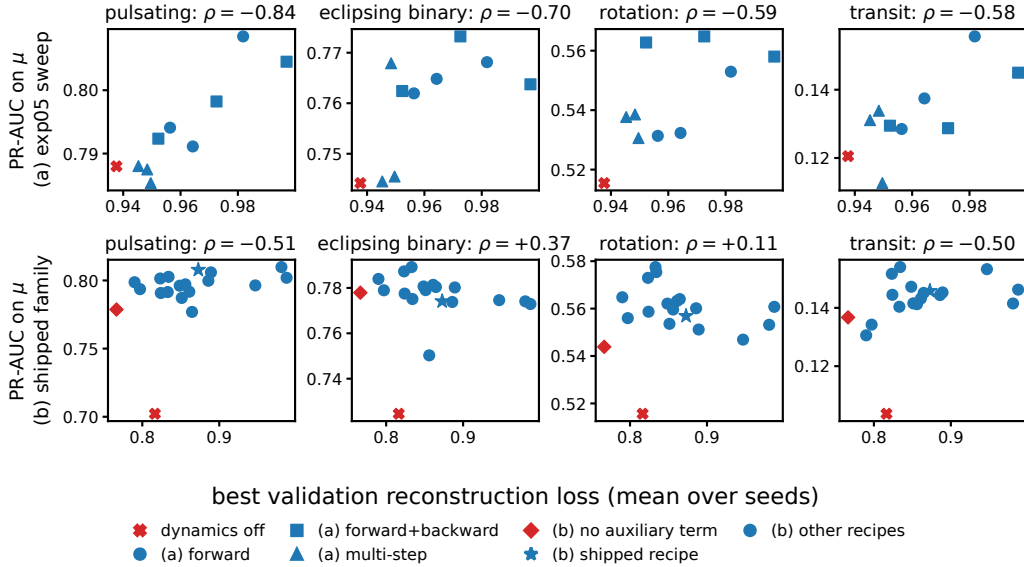


Figure 2: One point per pre-training recipe (mean over its seeds), one panel per task. (a) The exp05 sweep: ten recipes sharing corpus, architecture and loss terms, differing only in the latent-dynamics term (off, forward, forward+backward, multi-step, the last three at three weights), four seeds each. (b) The family the shipped encoder belongs to: 19 recipes on the final encoder that vary the form, weight and floor of the spectral auxiliary term, six seeds each; the diamond has no auxiliary term, the cross is the one dynamics-off recipe, the star is the shipped recipe. x : best post-warm-up validation reconstruction loss; y : PR-AUC of the fixed linear probe on μ ; ρ : Spearman's between negated loss and score.

Figure 2 is the measurement behind the model-selection paragraph of §4. In row (a) the loss-optimal recipe is the dynamics-free one (red cross) on all four tasks, and it is the worst or next-to-worst probe on each, so selecting on the loss costs 0.021–0.049 PR-AUC against the probe-optimal recipe. The trend is not carried by that one point: with it removed, ρ still runs from -0.43 to -0.87 . Those recipes are an earlier sweep than the shipped encoder (they carry an earlier spectral auxiliary term), so row (b) repeats the estimator on the family we ship: 19 recipes built on the final encoder that vary the form, weight and floor of the auxiliary term, six seeds each (two more recipes in which one seed collapsed are excluded; we never drop a seed). There the anti-correlation does not persist as such: ρ is -0.51 on pulsating and -0.50 on transit, $+0.37$ on eclipsing binaries and $+0.11$ on rotation. What persists is that the loss is not a selector: its optimum is again the recipe with the weakest competing term (the diamond, no auxiliary term at all), which is never the probe optimum, and trusting it costs 0.011–0.034 PR-AUC. So row (a) shows an anti-correlation on every task; in row (b) it survives on two tasks and reverses on two; in neither does the loss choose. Both are statements across recipes only: within one recipe, across its seeds, no label-free curve metric we tried (reconstruction, reconstruction plus auxiliary, active latent units) ranks the seeds, every such correlation sitting within ± 0.3 of zero.

C Paired bootstrap over test stars

The 2 SE bands in the body come from re-training and say nothing about whether the particular test stars favoured one arm. For every printed contrast we therefore resample each task's test set with replacement 1,000 times, rescore every arm on the identical resampled list from its saved per-star predictions (no model is retrained), average each arm over its seeds, and take the difference. Table 2 gives the 95% percentile interval of that difference and the share of resamples in which the fusion

arm is ahead; intervals that exclude zero are bold. The check tightens no verdict and reverses none: every row the body calls ahead has the fusion arm ahead in at least 65% of resamples. It does loosen the smallest tasks – rotation period (150 test stars) and RGB versus helium-burning (161) account for four of the ten rows where a resolved 2 SE verdict becomes an interval that touches zero.

task	fusion – features		fusion – C1		fusion – C2		fusion – dynamics-off fusion	
	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$
pulsating	[+0.012, +0.105]	0.99	[-0.008, +0.073]	0.94	[+0.045, +0.150]	1.00	[-0.023, +0.036]	0.69
eclipsing binary	[+0.005, +0.062]	0.99	[-0.024, +0.025]	0.52	[+0.022, +0.094]	1.00	[+0.004, +0.038]	0.99
rotation	[-0.026, +0.057]	0.77	[-0.056, +0.040]	0.38	[-0.023, +0.079]	0.87	[-0.005, +0.052]	0.94
transit	[-0.025, +0.065]	0.85	[+0.041, +0.124]	1.00	[+0.046, +0.174]	1.00	[-0.016, +0.035]	0.78
oscillating giants	[-0.010, +0.005]	0.21	[+0.028, +0.057]	1.00	[+0.021, +0.051]	1.00	[-0.002, +0.008]	0.87
solar-like osc.	[+0.044, +0.097]	1.00	[-0.037, +0.022]	0.34	[+0.079, +0.151]	1.00	[+0.025, +0.067]	1.00
RGB vs. HeB	[-0.087, -0.012]	0.00	[+0.099, +0.247]	1.00	[-0.126, +0.020]	0.08	[-0.037, +0.039]	0.52
$\nu_{\max}$	[+0.028, +0.043]	1.00	[+0.024, +0.045]	1.00	[+0.018, +0.047]	1.00	[+0.020, +0.031]	1.00
rotation period	[-0.038, +0.067]	0.65	[+0.054, +0.186]	1.00	[-0.053, +0.100]	0.69	[-0.021, +0.060]	0.83
flare	[+0.019, +0.078]	1.00	[-0.047, +0.029]	0.32	[+0.027, +0.128]	1.00	[-0.000, +0.035]	0.97

Table 2: Paired star-bootstrap of the four contrasts the body counts. Each entry is the 95% interval of (left arm – right arm) over 1,000 resamples of the test stars, seeds averaged, and the share of resamples with the difference above zero ($f_{>0}$; 1.00 means the left arm is ahead in every resample, 0.00 that it is behind in every resample). Bold: interval excludes zero.

D The 25 engineered features

All 25 features are computed from a star’s first segment: its windows concatenated into one flux series x of 4,096 or 5,120 cadences, already median-normalised by its median absolute deviation. The periodogram is the power spectrum of the mean-subtracted series, $P(f) = |\text{FFT}(x - \bar{x})|^2$ with the zero-frequency bin dropped, and $p(f) = P(f) / \sum_f P(f)$ its normalisation to unit sum. Frequencies are in cycles per day. f_1, f_2, f_3 are the three strongest local maxima of P , strongest first.

#	feature	definition
1–3	$\log_{10} f_1, \log_{10} f_2, \log_{10} f_3$	frequencies of the three strongest periodogram peaks
4–6	p_1, p_2, p_3	normalised power at each peak, $p_k = P(f_k) / \sum_f P(f)$
7–8	$p_2/p_1, p_3/p_1$	peak-power ratios
9	harmonic ratio	$P(2f_1)/P(f_1)$ at the bin nearest $2f_1$; 0 if $2f_1$ exceeds the Nyquist frequency
10–17	band powers	$\log_{10} \sum_{f \in B_j} p(f)$ for eight log-spaced bands B_j spanning 0.3–360 c/d
18	spectral entropy	$-\sum_f p(f) \ln p(f) / \ln N$, N the number of frequency bins
19	spectral centroid	$\log_{10} \sum_f f p(f)$
20–21	skew, kurtosis	sample skewness and excess kurtosis of x
22	point-to-point scatter ratio	$\text{std}(\Delta x) / \text{std}(x)$, Δ the first difference
23	depth	95th minus 5th percentile of x
24	MAD	median absolute deviation of x
25	IQR	inter-quartile range of x

Table 3: The engineered feature vector, modelled on the T²DA variability pipeline [Audenaert et al., 2021].